

- (a) The mean of the state is the Kalman filter estimate of the state.
 - (b) The covariance of the state is the covariance of the Kalman filter state estimate.
4. Every time that we get a measurement, we update the mean and covariance of the state. This is similar to what we did in Chapter 3 where we used measurements to recursively update our estimate of a constant.

In Section 5.1, we derive the equations of the discrete-time Kalman filter. This includes several different-looking, but mathematically equivalent forms. Various books and papers that deal with Kalman filters present the filter equations in ways that appear very different from one another. It is not always obvious, but these different formulations are actually mathematically equivalent, and we will see this in Section 5.1. (Sections 9.1, 10.5.1, and 11.1 also derive alternate but equivalent formulations of the Kalman filter equations.) In Section 5.2, we will examine some of the theoretical properties of the Kalman filter. One remarkable aspect of the Kalman filter is that it is optimal in several different senses, as we will see in Section 5.2. In Section 5.3, we will see how the Kalman filter equations can be written with a single time update equation. Section 5.4 presents a way to obtain a closed-form equation for the time-varying Kalman filter for a scalar time-invariant system, and a way to quickly compute the steady-state Kalman filter. Section 5.5 looks at some situations in which the Kalman filter is unstable or gives state estimates that are not close to the true state. We will also look at some ways that instability and divergence can be corrected in the Kalman filter.

5.1 DERIVATION OF THE DISCRETE-TIME KALMAN FILTER

Suppose we have a linear discrete-time system given as follows:

$$\begin{aligned}x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k\end{aligned}\tag{5.1}$$

The noise processes $\{w_k\}$ and $\{v_k\}$ are white, zero-mean, uncorrelated, and have known covariance matrices Q_k and R_k , respectively:

$$\begin{aligned}w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \\ E[w_k w_j^T] &= Q_k \delta_{k-j} \\ E[v_k v_j^T] &= R_k \delta_{k-j} \\ E[v_k w_j^T] &= 0\end{aligned}\tag{5.2}$$

where δ_{k-j} is the Kronecker delta function; that is, $\delta_{k-j} = 1$ if $k = j$, and $\delta_{k-j} = 0$ if $k \neq j$. Our goal is to estimate the state x_k based on our knowledge of the system dynamics and the availability of the noisy measurements $\{y_k\}$. The amount of information that is available to us for our state estimate varies depending on the particular problem that we are trying to solve. If we have all of the measurements up to and including time k available for use in our estimate of x_k , then we can form an *a posteriori* estimate, which we denote as $\hat{x}_k^+$. The “+” superscript denotes that

the estimate is *a posteriori*. One way to form the *a posteriori* state estimate is to compute the expected value of x_k conditioned on all of the measurements up to and including time k :

$$\hat{x}_k^+ = E[x_k | y_1, y_2, \dots, y_k] = \text{a posteriori estimate} \quad (5.3)$$

If we have all of the measurements before (but not including) time k available for use in our estimate of x_k , then we can form an *a priori* estimate, which we denote as $\hat{x}_k^-$. The “-” superscript denotes that the estimate is *a priori*. One way to form the *a priori* state estimate is to compute the expected value of x_k conditioned on all of the measurements before (but not including) time k :

$$\hat{x}_k^- = E[x_k | y_1, y_2, \dots, y_{k-1}] = \text{a priori estimate} \quad (5.4)$$

It is important to note that $\hat{x}_k^-$ and $\hat{x}_k^+$ are both estimates of the same quantity; they are both estimates of x_k . However, $\hat{x}_k^-$ is our estimate of x_k *before* the measurement y_k is taken into account, and $\hat{x}_k^+$ is our estimate of x_k *after* the measurement y_k is taken into account. We naturally expect $\hat{x}_k^+$ to be a better estimate than $\hat{x}_k^-$, because we use more information to compute $\hat{x}_k^+$:

$$\begin{aligned} \hat{x}_k^- &= \text{estimate of } x_k \text{ before we process the measurement at time } k \\ \hat{x}_k^+ &= \text{estimate of } x_k \text{ after we process the measurement at time } k \end{aligned} \quad (5.5)$$

If we have measurements after time k available for use in our estimate of x_k , then we can form a *smoothed* estimate. One way to form the smoothed state estimate is to compute the expected value of x_k conditioned on all of the measurements that are available:

$$\hat{x}_{k|k+N} = E[x_k | y_1, y_2, \dots, y_k, \dots, y_{k+N}] = \text{smoothed estimate} \quad (5.6)$$

where N is some positive integer whose value depends on the specific problem that is being solved. If we want to find the best prediction of x_k more than one time step ahead of the available measurements, then we can form a *predicted* estimate. One way to form the predicted state estimate is to compute the expected value of x_k conditioned on all of the measurements that are available:

$$\hat{x}_{k|M} = E[x_k | y_1, y_2, \dots, y_{k-M}] = \text{predicted estimate} \quad (5.7)$$

where M is some positive integer whose value depends on the specific problem that is being solved. The relationship between the *a posteriori*, *a priori*, smoothed, and predicted state estimates is depicted in Figure 5.1.

In the notation that follows, we use $\hat{x}_0^+$ to denote our initial estimate of x_0 before any measurements are available. The first measurement is taken at time $k = 1$. Since we do not have any measurements available to estimate x_0 , it is reasonable to form $\hat{x}_0^+$ as the expected value of the initial state x_0 :

$$\hat{x}_0^+ = E(x_0) \quad (5.8)$$

We use the term P_k to denote the covariance of the estimation error. P_k^- denotes the covariance of the estimation error of $\hat{x}_k^-$, and P_k^+ denotes the covariance of the estimation error of $\hat{x}_k^+$:

$$\begin{aligned} P_k^- &= E[(x_k - \hat{x}_k^-)(x_k - \hat{x}_k^-)^T] \\ P_k^+ &= E[(x_k - \hat{x}_k^+)(x_k - \hat{x}_k^+)^T] \end{aligned} \quad (5.9)$$

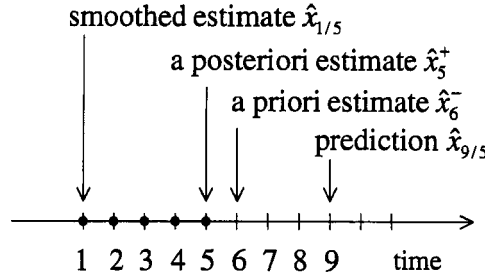


Figure 5.1 Time line showing the relationship between the *a posteriori*, *a priori*, smoothed, and predicted state estimates. In this figure, we suppose that we have received measurements at times up to and including $k = 5$. An estimate of the state at $k < 5$ is called a smoothed estimate. An estimate of the state at $k = 5$ is called the *a posteriori* estimate. An estimate of the state at $k = 6$ is called the *a priori* estimate. An estimate of the state at $k > 6$ is called the prediction.

These relationships are depicted in Figure 5.2. The figure shows that after we process the measurement at time $(k-1)$, we have an estimate of x_{k-1} (denoted $\hat{x}_{k-1}^+$) and the covariance of that estimate (denoted P_{k-1}^+). When time k arrives, before we process the measurement at time k we compute an estimate of x_k (denoted $\hat{x}_k^-$) and the covariance of that estimate (denoted P_k^-). Then we process the measurement at time k to refine our estimate of x_k . The resulting estimate of x_k is denoted $\hat{x}_k^+$, and its covariance is denoted P_k^+ .

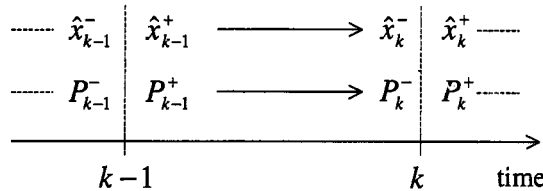


Figure 5.2 Timeline showing *a priori* and *a posteriori* state estimates and estimation-error covariances.

We begin the estimation process with $\hat{x}_0^+$, our best estimate of the initial state x_0 . Given $\hat{x}_0^+$, how should we compute $\hat{x}_1^-$? We want to set $\hat{x}_1^- = E(x_1)$. But note that $\hat{x}_0^+ = E(x_0)$, and recall from Equation (4.2) how the mean of x propagates with time: $\bar{x}_k = F_{k-1}\bar{x}_{k-1} + G_{k-1}u_{k-1}$. We therefore obtain

$$\hat{x}_1^- = F_0\hat{x}_0^+ + G_0u_0 \quad (5.10)$$

This is a specific equation that shows how to obtain $\hat{x}_1^-$ from $\hat{x}_0^+$. However, the reasoning can be extended to obtain the following more general equation:

$$\hat{x}_k^- = F_{k-1}\hat{x}_{k-1}^+ + G_{k-1}u_{k-1} \quad (5.11)$$

This is called the time update equation for $\hat{x}$. From time $(k-1)^+$ to time k^- , the state estimate propagates the same way that the mean of the state propagates. This makes sense intuitively. We do not have any additional measurements available to

help us update our state estimate between time $(k-1)^+$ and time k^- , so we should just update the state estimate based on our knowledge of the system dynamics.

Next we need to compute the time update equation for P , the covariance of the state estimation error. We begin with P_0^+ , which is the covariance of our initial estimate of x_0 . If we know the initial state perfectly, then $P_0^+ = 0$. If we have absolutely no idea of the value of x_0 , then $P_0^+ = \infty I$. In general, P_0^+ represents the uncertainty in our initial estimate of x_0 :

$$\begin{aligned} P_0^+ &= E[(x_0 - \bar{x}_0)(x_0 - \bar{x}_0)^T] \\ &= E[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T] \end{aligned} \quad (5.12)$$

Given P_0^+ , how can we compute P_1^- ? Recall from Equation (4.4) how the covariance of the state of a linear discrete-time system propagates with time: $P_k = F_{k-1}P_{k-1}F_{k-1}^T + Q_{k-1}$. We therefore obtain

$$P_1^- = F_0P_0^+F_0^T + Q_0 \quad (5.13)$$

This is a specific equation that shows how to obtain P_1^- from P_0^+ . However, the reasoning can be extended to obtain the following more general equation:

$$P_k^- = F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \quad (5.14)$$

This is called the time-update equation for P .

We have derived the time-update equations for $\hat{x}$ and P . Now we need to derive the measurement-update equations for $\hat{x}$ and P . Given $\hat{x}_k^-$, how should we compute $\hat{x}_k^+$? The quantity $\hat{x}_k^-$ is an estimate of x_k , and the quantity $\hat{x}_k^+$ is also an estimate of x_k . The only difference between $\hat{x}_k^-$ and $\hat{x}_k^+$ is that $\hat{x}_k^+$ takes the measurement y_k into account. Recall from the recursive least squares development in Section 3.3 that the availability of the measurement y_k changes the estimate of a constant x as follows:

$$\begin{aligned} K_k &= P_{k-1}H_k^T(H_kP_{k-1}H_k^T + R_k)^{-1} \\ &= P_kH_k^TR_k^{-1} \\ \hat{x}_k &= \hat{x}_{k-1} + K_k(y_k - H_k\hat{x}_{k-1}) \\ P_k &= (I - K_kH_k)P_{k-1}(I - K_kH_k)^T + K_kR_kK_k^T \\ &= (P_{k-1}^{-1} + H_k^TR_k^{-1}H_k)^{-1} \\ &= (I - K_kH_k)P_{k-1} \end{aligned} \quad (5.15)$$

where $\hat{x}_{k-1}$ and P_{k-1} are the estimate and its covariance *before* the measurement y_k is processed, and $\hat{x}_k$ and P_k are the estimate and its covariance *after* the measurement y_k is processed. In this chapter, $\hat{x}_k^-$ and P_k^- are the estimate and its covariance before the measurement y_k is processed, and $\hat{x}_k^+$ and P_k^+ are the estimate and its covariance after the measurement y_k is processed. These relationships are shown in Table 5.1.¹

We can now generalize from the formulas for the estimation of a constant in Section 3.3, to the measurement update equations required in this section. In

¹We need to use minus and plus superscripts on $\hat{x}_k$ and P_k in order to distinguish between quantities before y_k is taken into account, and quantities after y_k is taken into account. In Chapter 3, we did not need superscripts because x was a constant.

Table 5.1 Relationships between estimates and covariances in Sections 3.3 and 5.1

Section 3.3 Least squares estimation	Section 5.1 Kalman filtering
$\hat{x}_{k-1}$ = estimate before y_k is processed	$\hat{x}_k^-$ = <i>a priori</i> estimate
P_{k-1} = covariance before y_k is processed	P_k^- = <i>a priori</i> covariance
$\hat{x}_k$ = estimate after y_k is processed	$\hat{x}_k^+$ = <i>a posteriori</i> estimate
P_k = covariance after y_k is processed	P_k^+ = <i>a posteriori</i> covariance

Equation (5.15), we replace $\hat{x}_{k-1}$ with $\hat{x}_k^-$, we replace P_{k-1} with P_k^- , we replace $\hat{x}_k$ with $\hat{x}_k^+$, and we replace P_k with P_k^+ . This results in

$$\begin{aligned}
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \\
&= P_k^+ H_k^T R_k^{-1} \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-) \\
P_k^+ &= (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k R_k K_k^T \\
&= [(P_k^-)^{-1} + H_k^T R_k^{-1} H_k]^{-1} \\
&= (I - K_k H_k) P_k^-
\end{aligned} \tag{5.16}$$

These are the measurement-update equations for $\hat{x}_k$ and P_k . The matrix K_k in the above equations is called the Kalman filter gain.

The discrete-time Kalman filter

Here we summarize the discrete-time Kalman filter by combining the above equations into a single algorithm.

1. The dynamic system is given by the following equations:

$$\begin{aligned}
x_k &= F_{k-1} x_{k-1} + G_{k-1} u_{k-1} + w_{k-1} \\
y_k &= H_k x_k + v_k \\
E(w_k w_j^T) &= Q_k \delta_{k-j} \\
E(v_k v_j^T) &= R_k \delta_{k-j} \\
E(w_k v_j^T) &= 0
\end{aligned} \tag{5.17}$$

2. The Kalman filter is initialized as follows:

$$\begin{aligned}
\hat{x}_0^+ &= E(x_0) \\
P_0^+ &= E[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T]
\end{aligned} \tag{5.18}$$

3. The Kalman filter is given by the following equations, which are computed for each time step $k = 1, 2, \dots$:

$$\begin{aligned}
P_k^- &= F_{k-1} P_{k-1}^+ F_{k-1}^T + Q_{k-1} \\
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1}
\end{aligned}$$